

Separating Two Things Composite Modeling Provides: Knowing the Outer Function from Observing the Intermediates

Abstract

Composite Bayesian optimization gives a surrogate two advantages at once: it is told the algebraic form of the outer function, and it gets to see the intermediate quantities that function consumes. Every comparison in our benchmark study changes both together, so none of them can say which advantage produced the gain. This note designs, implements and validates a third surrogate that receives exactly one of the two, which splits the composite gain into its two parts. Part I is a short summary of the whole idea. The remaining parts give the full detail: the reasoning behind the decomposition, the synthetic problems, the inference procedure for the new surrogate and the three defects found in it, what is measured, and what has and has not been verified, and the measured results.

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Part I

Overview

1 The short version

The setting. Many expensive objectives factor as $f(x) = g(h(x))$, where $h : \mathcal{X} \rightarrow \mathbb{R}^k$ is an expensive black box (a simulator, an experiment) and $g : \mathbb{R}^k \rightarrow \mathbb{R}$ is a known, cheap function written down by whoever posed the problem. A *direct* surrogate fits a Gaussian process to f and ignores the factorization. A *composite* surrogate fits Gaussian processes to h and pushes their posterior samples through g . Our benchmark study found the composite surrogate better on most problems, by margins running from 121% down to nothing.

The problem with that comparison. Going from direct to composite changes two things simultaneously:

1. **knowledge of g** — the surrogate is handed the outer function and never has to learn it from data;
2. **observation of h** — each expensive evaluation now returns k numbers instead of one.

Both are bundled into every result we have. So when the composite surrogate wins by 121% on DTLZ2 we cannot say which advantage did the work, and when it wins by nothing on RGB we cannot say which one failed.

The fix. Add a third surrogate that gets advantage (1) and not (2): it knows g exactly, holds the same prior belief about h that the composite surrogate holds, but only ever observes the scalar y . Call it the *latent* arm, because h is never seen and must be inferred. Then

$$\text{latent} - \text{direct} = \text{the value of knowing } g, \tag{1}$$

$$\text{composite} - \text{latent} = \text{the value of observing } h, \tag{2}$$

and the two sum exactly to the composite gain we already report.

What governs the split. Observing h can only help to the extent that $y = g(h)$ fails to pin h down. That makes invertibility of g the controlling property, and it yields two extreme cases we can check against theory rather than intuition:

- **g injective** $\Rightarrow h = g^{-1}(y)$, so the latent and composite arms see identical information and are the same model. Observing h is worth *exactly zero*.
- **g linear** $\Rightarrow$ a linear combination of independent GPs is itself a GP, so the latent arm collapses into a direct GP on y . Knowing g is worth *approximately zero*. (Truss and RCM40, our two weakest results, both have linear outer maps.)

Between those, the information g destroys is the *fiber* $g^{-1}(y)$: a pair of points for z^2 , several for $\cos$, a half-line for a clipped map, a circle for a norm, a hyperbola for a product.

The experiment. Nineteen outer maps applied to intermediates drawn from the same GP family the surrogates assume, so the inner function is never a confound. Single objective, $d = 6$, 5 initial points plus 40 adaptive evaluations, all three arms sharing the initial design and the acquisition function so the surrogate is the only difference. Four independent draws of the inner function, five initial designs each.

The hard part. Arms 1 and 2 have closed-form posteriors — ordinary GP regression. The latent arm does not: it must infer quantities it never observed through a nonlinear g , which is Markov chain Monte Carlo rather than a formula. Every difficulty in this project has been there, and three genuine defects were found and fixed (Section 14).

Why we can believe the result. Three of the nineteen conditions have answers known in advance — the identity map, an injective map, and a linear map — and are included solely so the machinery can be graded where grading is possible. That is what caught the defects: on the identity map “observing h ” must be exactly zero, and the code reported $+0.956$. After the fixes it reports -0.096 ± 0.175 , and the two other controls pass as well.

2 Reading guide

Part II gives the argument in full, including the worked cases behind each prediction. Part III specifies the synthetic problems and explains the two that had to be redesigned. Part IV is the inference procedure for the latent arm, the three defects and their fixes. Part V defines every reported quantity, including the diagnostics. Part VI is the protocol and the validation evidence. Part VII states the limitations. The appendix maps each piece of the design onto the code that implements it.

Part II

The idea in detail

3 The composite setting

Fix a single objective $f(x) = g(h(x))$ on $\mathcal{X} = [0, 1]^d$, with $h : \mathcal{X} \rightarrow \mathbb{R}^k$ unknown and $g : \mathbb{R}^k \rightarrow \mathbb{R}$ known in closed form. After n evaluations write $X = (x_1, \dots, x_n)$, $\mathbf{y} = (y_1, \dots, y_n)$ with $y_i = f(x_i) + \varepsilon_i$, and $\mathbf{h}(X) = (h(x_1), \dots, h(x_n)) \in \mathbb{R}^{n \times k}$ for the matrix of intermediates.

A surrogate’s job at each iteration is to produce a predictive distribution over $f(x')$ at candidate points x' , which an acquisition function turns into the next evaluation. The three arms differ only in how that predictive is built.

4 The confound, stated precisely

Write I_{direct} , I_{comp} for the information each surrogate conditions on:

$$I_{\text{direct}} = (\mathbf{y}), \quad g \text{ unknown}, \quad (3)$$

$$I_{\text{comp}} = (\mathbf{h}(X)), \quad g \text{ known}. \quad (4)$$

These differ in two coordinates at once. Any performance gap between them is a sum of two effects with no way to separate them from the two numbers alone. The latent arm supplies the missing corner of the 2×2 table:

	g unknown	g known
observes y only	arm 1 (direct)	arm 3 (latent)
observes h	— (incoherent)	arm 2 (composite)

The empty cell is incoherent: a surrogate that observes h but does not know g cannot predict f at all. So three arms are the most that can be instantiated, and three is enough — the two available differences are exactly the two effects, and they telescope.

5 The three arms

Arm	Observes	Knows g	Surrogate
1. Direct	$y_i = f(x_i)$	no	$f \sim \mathcal{GP}$ fit to y , hyperparameters by MLE
2. Composite	$h(x_i)$	yes	$h_j \sim \mathcal{GP}$ conditioned on observed h_j , then apply g
3. Latent	$y_i = f(x_i)$	yes	$h_j \sim \mathcal{GP}$ <i>a priori</i> , h inferred from y

5.1 What the latent arm computes

Its posterior over the unobserved intermediates is Bayes' rule and nothing more:

$$p(\mathbf{h}(X) \mid \mathbf{y}) \propto \underbrace{\prod_{i=1}^n \mathcal{N}(y_i; g(h(x_i)), \sigma^2)}_{\text{likelihood: does this } h \text{ explain } y?} \cdot \underbrace{\prod_{j=1}^k \mathcal{N}(h_j(X); m_j, v_j K)}_{\text{GP prior: is this } h \text{ a plausible function?}}, \quad (5)$$

where $K_{ii'} = k_{\text{RBF}}(x_i, x_{i'})$ and (m_j, v_j) are the prior mean and variance of the j th intermediate.

Prediction is then exact *given* the latents, because $h(x') \mid \mathbf{h}(X)$ is an ordinary GP conditional. So the arm-3 predictive is a **mixture of Gaussian processes**, one component per posterior sample of $\mathbf{h}(X)$:

$$p(f(x') \mid \mathbf{y}) = \int p(g(h(x')) \mid \mathbf{h}(X)) p(\mathbf{h}(X) \mid \mathbf{y}) d\mathbf{h}(X) \approx \frac{1}{S} \sum_{s=1}^S p(g(h(x')) \mid \mathbf{h}^{(s)}). \quad (6)$$

That mixture is the whole point. A direct GP on y is a single Gaussian and cannot represent multimodality; if two very different intermediate functions both explain the data, the latent arm carries both hypotheses forward and its predictive is genuinely bimodal.

5.2 The prior, and what “sharing” it means

A GP prior here is three numbers per intermediate: a mean m_j , a variance v_j , and a lengthscale ℓ shared across coordinates. Everything else the GP does follows from those plus the data.

Arms 2 and 3 are constructed from the **same** prior object — same (m_j, v_j, ℓ) , same conditioning code — and it is not an estimate: it is the recipe the inner functions were actually generated from (Section 7). The only difference between the two arms is therefore what gets conditioned on: arm 2 conditions on $\mathbf{h}(X)$ directly, arm 3 must first infer it from $\mathbf{y}$. That is deliberate. Had arm 3 been given a different lengthscale or variance, the gap between arms 2 and 3 would blend “better prior” with “better observations” and measure neither.

Arm 1 cannot share it and does not: it models y , and for a general g the function $g(h(\cdot))$ is not a GP, so there is no true prior to hand it. It estimates its own by maximum likelihood. Section 18 deals with the asymmetry this creates.

5.3 The accounting identity

With R_a the performance of arm a on a common metric,

$$\underbrace{(R_1 - R_3)}_{\text{knowing } g} + \underbrace{(R_3 - R_2)}_{\text{observing } h} = \underbrace{(R_1 - R_2)}_{\text{composite gain we already report}}. \quad (7)$$

The identity is trivially true, which is a feature: the right-hand side is a number we have already published for each benchmark, so the decomposition is guaranteed to add up and any failure to do so is a bug, not a finding.

6 Why the decomposition should be informative

6.1 Invertible g : observing h is worth nothing

If g is injective and observations are noise-free, $h(x_i) = g^{-1}(y_i)$ exactly. Arm 3 recovers the intermediates from the objective alone, so arms 2 and 3 condition on identical information and are literally the same model. Any composite advantage on such a problem is attributable entirely to knowing g .

The exponential warping family $g(z) = \exp(\beta z)$ is of this type. Knowing g is still worth something — it tells the surrogate that $y > 0$, and how the scale is warped — but the extra numbers are redundant.

Two of these maps are in the run specifically as a sanity check: if the code reports anything other than exactly zero for “observing h ” on $\exp$ or $\sinh$, it is broken.

6.2 Linear g : knowing g is worth nothing

If $g(\mathbf{y}) = \sum_j c_j y_j$ and the h_j are independent GPs, then $g(h(\cdot))$ is *itself* a GP with mean $\sum_j c_j m_j$ and kernel $(\sum_j c_j^2 v_j)k(\cdot, \cdot)$. Arm 3 therefore collapses to an ordinary GP fitted to y — which is arm 1 with a particular kernel. Up to hyperparameter estimation, knowing g buys nothing.

This is the sharpest prediction in the design, and it bears directly on the paper: Truss and RCM40 both have linear outer maps and both produced our weakest results. If the prediction holds, the explanation is not the vague “linear maps are easy” but something specific — on a linear map there is no knowledge to exploit, so whatever gain the composite surrogate shows must come from the extra observations alone.

6.3 Non-invertible g : both channels are live

Take the smallest interesting case, $g(z) = z^2$, noise-free. Then $h(x)$ is known only up to sign:

$$p(h(x') \mid g(h(x)) = y) = p(h(x') \mid h(x) \in \{+\sqrt{y}, -\sqrt{y}\}), \quad (8)$$

so the posterior is a two-component mixture after one observation, and a mixture over 2^n sign patterns after n , each pattern weighted by its plausibility under the GP prior. Crucially the weights are *not* uniform: a pattern that requires the function to jump between $+\sqrt{y}$ and $-\sqrt{y}$ at two nearby inputs is heavily penalized by the prior, so the data do constrain the signs jointly even though each observation alone does not. Observing h removes the ambiguity outright, which is what arm 2 measures.

But knowing g is worth something even with no observation of h at all. The range of g is $[0, \infty)$, so arm 3 assigns

$$P(f(x') = -1 \mid \mathcal{D}) = P(h(x')^2 = -1 \mid \mathcal{D}) = 0, \quad (9)$$

whereas a direct GP on y — a Gaussian — puts positive probability on values that can never occur. More generally, knowing g imposes the support, the curvature, and any kinks or saturation of the outer map, none of which a stationary GP on y can represent.

In general the information destroyed by g is the *fiber* $g^{-1}(y)$: a finite set for z^2 or $\cos$, a half-line for a clipped or rectified map, a circle for a Euclidean norm, a hyperbola for a product. This is the structure the synthetic maps in Part III are chosen to vary.

6.4 Predictions, stated before the run

g	Type	knowing g (arm 1 – arm 3)	observing h (arm 3 – arm 2)
z	identity	0	0
$\exp(\beta z), \sinh(\beta z)$	invertible	+	0 by construction
$(z - c)^2$	lossy, tunable	+	+ at $c = 0$, $\rightarrow 0$ as c grows
$ z $	lossy	+	+
$y_1 + y_2$	linear	≈ 0	+
norm, max, product, ratio	lossy	+	+
$-\max(z, 0)$	censors away from optimum	+	≈ 0
$\text{clip}(2y_1, -1, 1) + y_2$	censors at the optimum	+	+

The last two rows deserve emphasis because they are the design's one genuinely non-obvious prediction. Both maps destroy information by censoring, but $-\max(z, 0)$ is optimized at the largest h , where g is injective, so the destroyed information sits *away from* the optimum; the clipped pair is constructed so the censored coordinate matters at the optimum. If observing h pays off on the second and not the first, then what matters is not how much information g destroys but *where*.

Part III

The test functions

7 Design principle: the inner function must not be a confound

If the inner functions were arbitrary analytic expressions, arms 2 and 3 might win or lose because a GP happens to model those expressions well or badly, and we would learn nothing about g . So every primitive h_j is a fixed draw from the *same* RBF-GP family the surrogates assume, generated with random Fourier features:

$$h(x) = \sqrt{\frac{2}{M}} \sum_{m=1}^M a_m \cos(\omega_m^\top x + b_m), \quad a_m \sim \mathcal{N}(0, 1), \quad b_m \sim U(0, 2\pi), \quad \omega_m \sim \mathcal{N}(0, \ell^{-2}I), \quad (10)$$

with $M = 1024$ and $\ell = 0.5$. As $M \rightarrow \infty$ this converges to a draw from $\mathcal{GP}(0, k_{\text{RBF}})$; at $M = 1024$ it is close enough that the prior handed to arms 2 and 3 is correct rather than approximately correct.

The draw is standardized on a fixed Sobol reference set, so the prior on the standardized function is $\mathcal{GP}(-\mu/s, k_{\text{RBF}}/s^2)$ with μ, s the measured mean and standard deviation. Those two constants are exactly what arms 2 and 3 receive as (m_j, v_j) . The same draw is reused across every outer map, so switching g never changes the underlying function.

Why standardizing matters beyond scale. It puts the draw’s mean at zero, so h takes *both* signs over the domain. Without that, z^2 would be injective on the region actually visited and would destroy nothing — the sweep’s whole knob would be dead.

Objective normalization. Each final objective is mapped to roughly $[0, 1]$ using the 1st and 99th percentiles of $g(h(\cdot))$ on a fixed reference set, so outer maps do not differ merely in scale. The transform is affine and known, so it is folded into g for all three arms and changes nothing about who knows what.

8 Function draws versus seeds

Two independent sources of randomness must be varied, and they are not interchangeable.

The initial design. Which 5 Sobol points start the run. All three arms share it within a seed, which is what makes the comparison paired.

The function draw. Which random h was drawn in the first place.

Only the first was varied in the initial version of this study: the function-draw seed was fixed, so twenty “seeds” meant twenty initial designs on *one* sampled function, and the reported standard errors described variation in the starting points only. A conclusion could have been a property of that one draw with no way to tell from the numbers. The run now sweeps four draws with five designs each — identical total compute — and the collection step pools them, so the error bars cover both sources.

9 The maps

Nineteen conditions in four groups.

Controls (3). $g(z) = z$; $g(\mathbf{y}) = y_1 + y_2$; $g(z) = \exp(z)$. Identity checks the machinery end to end (both effects are zero). The linear map checks the sampler against an exactly computable posterior. The injective map checks that “observing h ” comes out at zero when theory says it must.

The sweep (5). $g_c(z) = (z - c)^2$ for $c \in \{0, 0.5, 1, 2, 3\}$ holds the functional form fixed and varies only how much ambiguity there is. At $c = 0$ the sign of h is lost wherever the draw crosses zero; at $c = 3$ the shifted argument almost never changes sign over the domain, so g is effectively injective. This is the cleanest available knob on “how much g discards,” and the central plot of the study is the “observing h ” gain against the measured U_{lost} along it.

Matched pairs (9). Each pair changes one property and holds the rest:

Pair	Shared	Isolates
z^2 vs. $ z $	both two-to-one	smoothness at the optimum
$\exp(\beta z)$ vs. $\sinh(\beta z)$	both injective, both smooth	whether a range constraint helps
$\cos(\omega z)$, $\omega \in \{1, 2, 4\}$	smooth, periodic	<i>how many</i> preimages
$-\max(z, 0)$ vs. clipped pair	both censor	<i>where</i> the censoring bites

Benchmark analogs (5). Two intermediates into one objective, each mirroring a real benchmark: $y_1 + y_2$ (Truss, RCM40), $y_1 y_2$ (DTLZ2), $\sqrt{y_1^2 + y_2^2}$ (Color), $\max(y_1, y_2)$ (Lake), and $u_1/(0.05 + u_1 + \lambda u_2)$ with $u = \text{sigmoid}(y)$ (RGB). These let the synthetic conclusions be carried back to the benchmark results.

10 Two maps that had to be redesigned

The first version of the censoring pair was degenerate, and the failure is instructive.

The symptom. On $g(z) = \max(z, 0)$ and $g(z) = \text{clip}(cz, -1, 1)$, all three arms reported a final $\log_{10}$ regret of exactly -8.000 , which is the clamp floor in the metric — i.e. zero regret for every arm on every seed.

The cause. Both maps are minimized on a flat region of positive measure. A bare $\max(z, 0)$ takes its minimum value 0 on the entire censored half $\{h \leq 0\}$, and a bare clip takes its minimum on the whole lower saturation region. The five-point Sobol initial design already contains such a point with high probability, so every arm starts at the optimum and the condition carries no signal about anything.

The fix. Both are now minimized at a unique point while censoring the same set:

$$g(z) = -\max(z, 0) \quad \text{optimum at } \max h, \text{ unique; censors } \{h \leq 0\} \quad (11)$$

$$g(\mathbf{y}) = \text{clip}(2y_1, -1, 1) + y_2 \quad \text{optimum is a genuine tradeoff} \quad (12)$$

The negation puts the optimum where g is injective, which is what turns this pair into the “where does the censoring bite” contrast of the previous section. The clipped version adds a second, uncensored intermediate so the objective is not flat, while the first coordinate stays censored inside saturation.

Verification. The median final regret of a five-point Sobol design alone rose from 0 to 1.02 for the rectified map and 0.62 for the clipped pair, i.e. there is now something to optimize. A four-seed pilot separated the arms as predicted: $U_{\text{lost}} = 0.088$ with “observing h ” at -0.045 ± 0.194 for the rectified map, against $U_{\text{lost}} = 0.459$ and $+2.811 \pm 1.295$ for the clipped pair.

Part IV

Inference for the latent arm

11 Why this needs a sampler at all

Arms 1 and 2 are ordinary GP regression: the posterior is Gaussian and available in closed form. Arm 3 is not, and the reason is structural rather than technical. Its likelihood in (5) depends on the latent $\mathbf{h}(X)$ through g , which is nonlinear; a Gaussian prior times a non-Gaussian likelihood is not Gaussian, so there is no formula. Asking “what would knowing g alone be worth?” necessarily means asking a surrogate to reason about quantities it never observed, and that is an inference problem the other two arms simply do not have.

The state being sampled is small — $n \times k$ numbers, at most $45 \times 2 = 90$ at our budget — but it is awkward: the posterior is multimodal by construction (that is the phenomenon under study), the likelihood is sharply peaked when σ is small, and several of the maps have kinks or flat regions so gradients are unavailable or uninformative.

12 Closed forms where they exist

Identity and injective maps. $h = g^{-1}(y)$ exactly, so arm 3 is implemented as arm 2. This is a shortcut, not a different model.

Linear maps. The posterior (5) is Gaussian and available analytically; arm 3 samples it directly.

Both shortcuts can be disabled with a flag so the general sampler runs on those maps too, which is how they are validated (Section 20).

13 The general sampler

Two move types alternate.

13.1 Elliptical slice sampling

Elliptical slice sampling [1] is the standard sampler for a GP prior with an arbitrary likelihood. Given the current state $\mathbf{h}$ it draws $\boldsymbol{\nu}$ from the prior, proposes

$$\mathbf{h}(\theta) = \mathbf{h} \cos \theta + \boldsymbol{\nu} \sin \theta, \tag{13}$$

and shrinks the bracket on θ until the likelihood threshold is met. The ellipse is constructed so that every point on it has the same prior density, which is why no acceptance ratio is needed. It requires no gradients — essential, because $|z|$, $\max$ and the clipped map are non-differentiable — and no step size to tune.

13.2 Level-set moves, and why they are necessary

Elliptical slice sampling alone is not enough. When σ is small the likelihood is sharply peaked on the constraint set $\{\mathbf{h} : g(h(x_i)) = y_i \forall i\}$, so a proposal must shrink to $\theta \sim \sigma/\text{sd}(h)$ to remain inside it. Movement *along* the constraint set is then a very slow random walk — and that set is precisely the ambiguity the experiment is about. A sampler that cannot traverse it will confidently report one branch and miss the multimodality entirely.

Those directions are known in closed form for each g . We therefore propose $h(x_i) \mapsto h'(x_i)$ with

$$g(h'(x_i)) = g(h(x_i)) \quad \text{exactly,} \quad (14)$$

from a symmetric proposal density. The likelihood ratio is then identically one and the Metropolis–Hastings acceptance probability reduces to the GP prior ratio alone. With $Q = K^{-1}$ and $r_j = h_j(X) - m_j$, changing coordinate i of latent j from α to β gives

$$\Delta \log p = -\frac{1}{2v_j} \left[Q_{ii}(\beta^2 - \alpha^2) + 2(\beta - \alpha)((Qr_j)_i - Q_{ii}\alpha) \right], \quad (15)$$

which is $O(n)$ per single-site move given Qr_j . Sites are visited in random scan order.

The moves are the symmetries of each map:

Map	Level-set move
$(z - c)^2, z $	reflection $h \mapsto 2c - h$
$\cos(\omega z)$	reflection through the nearest multiple of π/ω ; period shift
$-\max(z, 0)$	random walk confined to the censored region
clipped pair	random walk on the clipped coordinate within one saturation region
$\sqrt{y_1^2 + y_2^2}$	rotation of (h_1, h_2)
$y_1 y_2$	negation of both; reciprocal rescaling $(ch_1, h_2/c)$
$\max(y_1, y_2)$	coordinate swap; move the smaller coordinate
$y_1 + y_2$	shift $(h_1 + \delta, h_2 - \delta)$

Only the ratio map has no convenient exact symmetry and relies on elliptical slice sampling alone — noted as a limitation in Part VII.

14 Three defects, and what they teach

The first full run produced a table that looked entirely reasonable: smooth trends across the sweep, sensible signs everywhere, tight standard errors. It was wrong. The identity control read “observing h ” = $+0.956 \pm 0.274$ where the true value is exactly 0 by construction. Three separate defects contributed.

14.1 Defect 1: collapsed Monte Carlo pairing

The predictive samples were formed by giving every latent sample a *shared* noise draw. When the latent posterior is tight, samples then differ only through their nearly identical posterior means, so the effective sample size collapses to one. Fixed by drawing n independent noise vectors and pairing each with a latent sample cycled from the chain, matching the composite arm’s sample count so the two are comparable. This was a real bug but not the main one: the identity control barely moved.

14.2 Defect 2: a false lead

The next hypothesis was that σ was acting as a resolution floor. It was tested directly at $\sigma \in \{10^{-2}, 10^{-3}, 10^{-4}\}$, which gave +1.036, +0.883, +1.156 — no trend — and U_{lost} moved only from 0.0026 to 0.0006 when it should scale as σ^2 . That non-scaling was itself the clue: something other than the likelihood was setting the spread.

14.3 Defect 3: latents never reaching the constraint surface

Two compounding mistakes in the warm-start path.

1. Burn-in anneals σ downward so a chain started from the prior can find the data. That annealing was also being applied on warm starts, re-heating latents that were already correct; the short warm burn-in could not recover them.
2. A newly appended BO point was initialized at its GP conditional mean, which is **not** on the surface $\{g(h) = y\}$. As established above, elliptical slice sampling cannot travel to that surface once the likelihood is tight — its steps are $O(\sigma/\text{sd}(h))$.

So each BO iteration appended a point that stayed off the surface, and the error accumulated. Measured constraint residual across successive iterations of one run:

BO iteration	1	2	3	4	5	6
mean $\ g(h) - y\ $	0.00013	0.00435	0.02666	0.01903	0.04255	0.12144

The fix is in two parts: do not anneal on a warm start, and *project* new rows onto the constraint surface before sampling, by local gradient descent on $\|g(h) - y\|^2$ started from the GP conditional mean. Descent lands on the preimage nearest that mean, which keeps whichever branch the prior prefers; the level-set moves then explore the others. Projection is used only to initialize, so it biases nothing — the stationary distribution is still (5).

After the fix the identity control reads -0.096 ± 0.175 , $U_{\text{lost}} = 0$ exactly, and the residual is stable at ≈ 0.0015 across iterations.

14.4 Why this matters methodologically

None of the three defects was visible in the results. Each produced plausible numbers with plausible error bars, and the trends across the sweep were in the predicted direction throughout. What exposed them was a condition whose answer was known before the run. A study containing only the interesting maps would have produced a confident, wrong table — and the wrongness pointed in the direction of the hypothesis, which is the worst case.

The practical rule the design now follows: every arm is graded on at least one condition where the correct answer is available independently of the code being graded.

Part V

What is measured

15 Information destroyed by g

$$U_{\text{lost}} = \frac{1}{nk} \sum_{i=1}^n \sum_{j=1}^k \text{Var}(h_j(x_i) \mid \mathbf{y}). \quad (16)$$

This is the posterior variance of the intermediates *at the points already evaluated* — not at new points, where uncertainty would reflect ordinary GP extrapolation rather than anything about g . It is zero when g is injective and noise-free, and grows with the size of the fiber. It is the measured version of the quantity the whole design is organized around, and it is what the “observing h ” gain is plotted against.

16 Optimization performance

Each arm reports simple regret against a reference minimum f^* ; the headline number is the final

$$\log_{10}(\max(f_{\text{best}} - f^*, 10^{-8})). \quad (17)$$

The log is used because regret spans orders of magnitude across maps, and the clamp keeps a run that finds the optimum exactly from producing $-\infty$. It also means a reported -8.000 is a saturated value, which is how the degenerate maps of Section 10 were spotted.

Comparisons are paired: all three arms share an initial design within a seed, so the reported difference is the mean of the per-seed differences and its standard error is

$$\text{SE} = \sqrt{\frac{1}{S(S-1)} \sum_{s=1}^S (d_s - \bar{d})^2}, \quad d_s = R_a^{(s)} - R_b^{(s)}. \quad (18)$$

Pairing matters a great deal here: the variation between initial designs is larger than the effects being measured, and differencing within a seed removes it.

17 Sampler diagnostics, and what each one means

Three numbers are reported per condition. They answer different questions and the third is the one to trust.

$\hat{R}$ (Gelman–Rubin). Several chains are started independently; $\hat{R}$ compares between-chain spread to within-chain spread, and equals 1 when they are indistinguishable. The conventional threshold is 1.1. It is **unreliable here** whenever the posterior is nearly a point, because it divides by within-chain variance — which is exactly the injective and large- c cases.

Chain disagreement. The largest gap between any two chains’ predicted objective value at held-out points, in normalized objective units where the range is about 1. This is reported because it is the quantity BO actually cares about, and because it stays meaningful when $\hat{R}$ does not. Under ≈ 0.02 is comfortable.

Move acceptance. The fraction of proposed level-set moves accepted. Near zero means the sampler is confined to one branch — which is a *failure* if other branches are plausible and *correct* if they are not. It should therefore track U_{lost} , and on the sweep it does: both fall to zero together.

Critically, all diagnostics are computed on the predictive of f , not on $\mathbf{h}$ itself. Under an even g a chain that has globally flipped the sign of h represents the identical function; measuring on h would register that as catastrophic non-convergence when nothing is wrong.

18 A confound that is not yet resolved

Arms 2 and 3 are given the generator’s true kernel. Arm 1 must estimate its hyperparameters by maximum likelihood, because for a general g there is no true kernel for f . The “knowing g ” comparison therefore mixes the value of knowing g with the value of having a correct prior instead of one fitted to between 5 and 45 points.

The identity map measures the second effect *alone*, since knowing g is worth exactly nothing there. In the pre-fix run it read $+0.628 \pm 0.260$ — the same order of magnitude as most of the real effects, so this is not a minor correction. Two ways to handle it:

1. **Subtract the baseline**, assuming the hyperparameter advantage is roughly constant across maps. Cheap, and available from the current run, but the assumption is unverified.
2. **Re-run with maximum-likelihood hyperparameters in all three arms**, which removes the asymmetry rather than correcting for it. This is the honest version and is the main outstanding piece of work.

Until (2) is done, “knowing g ” should be read as an upper bound.

Part VI

Protocol and validation

19 Protocol

Single objective on $[0, 1]^6$ with $\ell = 0.5$; $n_0 = 5$ Sobol points and 40 adaptive evaluations, matching Experiment 1 of the benchmark study; four function draws $\times$ five initial designs per condition, pooled. All three arms share the initial design, the acquisition function, the candidate pool and the budget, so the surrogate is the only thing that differs.

The acquisition is Monte Carlo LogEI in a numerically stable form,

$$\log \text{EI}(x') \approx \log \left[\frac{1}{N} \sum_s \tau \text{softplus}((f_{\text{best}} - f_s(x'))/\tau) \right], \quad (19)$$

evaluated by `logsumexp` over $N = 256$ samples with $\tau = 10^{-3}$, on a 2048-point Sobol pool with local refinement of the best four candidates. The softplus smoothing matters for arm 3 specifically: with a mixture predictive the hard hinge $\max(0, \cdot)$ makes the acquisition surface non-smooth in a way that interacts badly with local refinement.

Runtime: about 2.5 minutes per (map, draw) single-threaded, so the full 19×4 grid is roughly 20–30 minutes on 16 workers. Single-threaded is deliberate — the matrices are at most 45×45

and BLAS threading measurably does not help at that size, so many one-thread jobs beat few many-thread ones.

20 Validation

Three checks, each on a condition whose answer is available independently. They are the reason the defects of Section 14 were caught, and they are run before every experiment.

20.1 Check 1: against an exactly computable posterior

On the linear map with $n = 20$ and the shortcut disabled, the posterior is Gaussian in closed form, so the sampler can be graded directly.

Quantity	Measured	Reading
$ \text{mean} - \text{exact mean} / \text{exact sd}$	median 0.069, max 0.364	off by 7% of a posterior sd
sampled sd / exact sd	median 0.968, range [0.84, 1.09]	3% too narrow on average
move acceptance	0.79	branches are being explored
$\hat{R}$ / chain disagreement	1.150 / 0.0085	chains agree to 0.85% of the range

The second row is the one that matters most: too-narrow posteriors make the surrogate overconfident, which stops it exploring. $\hat{R}$ exceeds 1.1 here, which is the case discussed above where it is not informative — the disagreement figure is the reliable one.

20.2 Check 2: calibration against known truth

On $\exp(z)$ with the shortcut disabled, the true h is recoverable, so we can ask whether the sampler’s error bars contain it. If calibrated, $z = |\text{posterior mean} - \text{true } h| / \text{posterior sd}$ behaves like $|\mathcal{N}(0, 1)|$: median ≈ 0.67 , about 95% below 2.

Measured: median 0.19, max 1.10, **100%** within two standard deviations. The posteriors are therefore somewhat *wider* than necessary, which is the safe direction. Before the fixes this read 95% with a largest raw error of 0.620; it is now 100% with 0.386.

The largest remaining error sits at $h = -1.96$, where $\exp(h) \approx 0.14$ and the slope of g falls below the noise level $\sigma = 10^{-2}$. This is correct behavior and a substantive caveat: **with observation noise, an injective g still destroys information wherever it is flat**, so “arm 2 equals arm 3 for injective g ” holds exactly only in the noise-free limit. The same map reports $U_{\text{lost}} = 2.4 \times 10^{-2}$ for this reason, rather than 0.

20.3 Check 3: does the designed knob actually turn?

Map	U_{lost}	$\hat{R}$	Chain disagreement	Move acceptance
$(z - 0)^2$	0.8703	1.013	0.0272	0.68
$(z - 0.5)^2$	1.0789	1.014	0.0331	0.54
$(z - 1)^2$	0.5154	1.129	0.0356	0.32
$(z - 2)^2$	0.1075	1.090	0.0261	0.07
$(z - 3)^2$	0.0038	1.226	0.0135	0.00
$ z $	0.8812	1.009	0.0356	0.66

Information destroyed falls by more than two orders of magnitude across the sweep, and move acceptance falls with it: by $c = 3$ the reflected proposal is so implausible under the prior that it is never accepted, which is the correct behavior when no ambiguity remains. Two independent measurements of the same collapse.

One blemish, recorded honestly. $c = 0.5$ reads *higher* than $c = 0$ (1.079 vs. 0.870), so U_{lost} is not monotone along the sweep. The automated check only tests the endpoints, so it passes. The likely explanation is that at $c = 0.5$ both preimages are still live but now asymmetric, giving a wide lopsided two-bump posterior whose variance genuinely exceeds that of the symmetric case; variance is not a monotone function of “how much ambiguity” when the modes are unequally weighted. Whether it survives pooling over function draws is an open question, and if it does, U_{lost} should be supplemented by a mode-count or entropy measure.

Chain disagreement on the sweep. It reads 0.026–0.036, above the 0.02 guideline, on exactly the maps whose posteriors are genuinely multimodal. This is not a defect — it is real posterior mass being reported — but it means the sweep’s numbers carry extra Monte Carlo noise on top of the seed variation. Pooling over function draws mitigates it, since draws average over which modes happen to matter.

21 Results

Nineteen conditions, four function draws, five initial designs each: 380 BO runs per arm. Everything below is pooled over the twenty (draw, seed) pairs with the paired standard error of Section V.

21.1 How to read the numbers

The reported quantity is a difference of final $\log_{10}$ regrets, so it converts to a multiplicative factor: an “observing h ” value of d means the composite arm ends with 10^d times lower regret than the latent arm. A value of 1 is a tenfold improvement; a value of 0 means the two arms are indistinguishable.

The bias correction. On *identity* with the shortcut disabled — so the general sampler runs on a problem where h is exactly recoverable and $U_{\text{lost}} = 0.0016$ — “observing h ” measures $+0.409 \pm 0.133$ where the true value is 0. The latent arm therefore carries a handicap unrelated to information, discussed in Section 21.6. Every map whose latent arm invokes the sampler is reported both raw and with 0.409 subtracted. The three maps that use the closed-form shortcut need no correction: there the latent arm *is* the composite arm and the measurement is exact.

21.2 What each condition tests

Map	g	What g destroys	Hypothesis
<i>Controls — answers known before the run</i>			
identity	z	nothing	both effects exactly 0
exp_b1	e^z	nothing (monotone, positive)	observing h exactly 0
sinh_b1	$\sinh z$	nothing (monotone, unbounded)	observing h exactly 0
sum	$y_1 + y_2$	how the sum splits	knowing $g \approx 0$
<i>The sweep — one form, information loss tuned by c</i>			
shiftsq_c0	$(z - 0)^2$	sign of z	largest gain
shiftsq_c0.5	$(z - 0.5)^2$	sign of $z - 0.5$	
shiftsq_c1	$(z - 1)^2$	sign of $z - 1$	gain falls
shiftsq_c2	$(z - 2)^2$	sign of $z - 2$ (rarely ambiguous)	with c
shiftsq_c3	$(z - 3)^2$	almost nothing	gain $\rightarrow 0$
<i>Matched pairs — one property changed at a time</i>			
abs	$ z $	sign of z , with a kink	as shiftsq_c0; kink is the only difference
cos_w1	$\cos z$	which of several preimages	gain grows with the
cos_w2	$\cos 2z$	which of more preimages	number of preimages
cos_w4	$\cos 4z$	which of many preimages	
relu	$-\max(z, 0)$	everything below 0, away from the optimum	gain ≈ 0
clip_c2	$\text{clip}(2y_1, -1, 1) + y_2$	the clipped coordinate at the optimum	gain large
<i>Benchmark analogs</i>			
sum	$y_1 + y_2$	how the sum splits	Truss, RCM40
product	$y_1 y_2$	how it factors, and sign	DTLZ2
norm	$\sqrt{y_1^2 + y_2^2}$	direction	Color
max	$\max(y_1, y_2)$	the smaller value	Lake
ratio	$\frac{u_1}{0.05 + u_1 + u_2}, u = \text{sigmoid}(y)$	scale	RGB

21.3 Measurements

Map	U_{lost}	observing h			knowing g	Verdict
		raw	corrected	factor	raw	
identity	0.000	−0.000	exact	1.0×	+0.409	as predicted
exp_b1	0.000	−0.000	exact	1.0×	+0.625	as predicted
sinh_b1	0.000	−0.000	exact	1.0×	+0.444	as predicted
shiftsq_c0	0.280	+1.899	+1.490	31×	+0.115	sweep runs backwards
shiftsq_c0.5	0.283	+1.688	+1.279	19×	+0.625	
shiftsq_c1	0.283	+2.303	+1.894	78×	−0.100	
shiftsq_c2	0.189	+3.159	+2.750	563×	−0.726	
shiftsq_c3	0.232	+2.923	+2.514	326×	−1.176	
abs	0.474	+0.352	−0.057	0.9×	−0.006	unlike shiftsq_c0
cos_w1	2.434	+2.861	+2.452	283×	−0.793	not monotone in ω
cos_w2	1.284	+1.471	+1.062	12×	+0.702	
cos_w4	1.161	+1.487	+1.078	12×	−0.797	
relu	0.075	+0.343	−0.066	0.9×	+0.201	
clip_c2	0.580	+1.517	+1.108	13×	+0.281	as predicted
sum	0.702	+0.229	−0.180	0.7×	+0.117	as predicted
product	1.640	+0.622	+0.213	1.6×	−0.411	
norm	0.426	+0.314	−0.095	0.8×	−0.024	
max	0.358	+0.548	+0.139	1.4×	+0.067	
ratio	0.267	+0.526	+0.117	1.3×	−0.418	

Paired standard errors run from 0.06 to 0.48; the full set is in `run.py --collect`. “Factor” is $10^{\text{corrected}}$, the ratio of final regrets.

21.4 What held

The invertible controls are exact. $−0.000 \pm 0.000$ on all three. When g is injective the latent arm recovers h from y and becomes the composite arm, and the code reproduces that identity to machine precision. This is the check that failed at +0.956 before the defects of Section 14 were fixed.

Knowing g is worth nothing on a linear map. $+0.117 \pm 0.340$ on `sum`, consistent with zero. The prediction was that a linear combination of independent GPs is itself a GP, so the latent arm collapses to a direct GP on y with a particular kernel. It does. This is the prediction that bears on the paper: Truss and RCM40 have linear outer maps, so whatever composite gain they show cannot come from knowing g .

Where the information is destroyed matters more than how much. This is the substantive result. `relu` and `clip_c2` both censor, and `relu` is the *more* destructive of the two by construction — it collapses an entire half-line to a point. Yet:

Map	Censoring sits	U_{lost}	observing h (corrected)
<code>relu</code>	away from the optimum	0.075	$−0.066 \pm 0.209$
<code>clip_c2</code>	at the optimum	0.580	$+1.108 \pm 0.421$

The rest of the table says the same thing. Sorting by U_{lost} does not sort by gain: **product** destroys the most information of any analog ($U_{\text{lost}} = 1.64$) and pays +0.213; **sum** destroys 0.702 and pays -0.180; **abs** destroys 0.474 and pays -0.057. Whether the ambiguity falls where the optimizer is looking is what separates them.

If this holds it sharpens the original hypothesis considerably. The quantity to compute for a new problem is not “is g invertible” but “is g invertible *near the Pareto front*”.

21.5 What did not hold: the sweep

The sweep was designed as the cleanest experiment in the set: one functional form, one parameter, information loss falling monotonically as c grows. The gain from observing h should have fallen with it. It rises — +1.49, +1.28, +1.89, +2.75, +2.51 — and the largest gains occur where g is *most* nearly injective.

U_{lost} explains why: it is flat at 0.19–0.28 across the whole sweep, where it should fall by two orders of magnitude. The knob does turn on the initial design — median U_{lost} at iteration 0 is 1.026, 0.869, 0.104, 0.013 for $c = 0, 1, 2, 3$, which is exactly the designed behavior — and then, over BO iterations, it falls for small c and *rises* for large c . More data cannot increase uncertainty about points already observed.

A second symptom points the same way: “knowing g ” goes negative at $c \geq 1$, reaching -1.176. The latent arm holds the generator’s true kernel and knows g exactly; losing to a direct GP by a factor of fifteen in final regret is not a null result.

Three candidate explanations were tested and two eliminated:

Hypothesis	Test	Outcome
Sampler budget too small	4 chains / burn 800 / keep 1500 vs. the loop’s 2/150/200	ruled out: U_{lost} un
Warm starts accumulate error	cold fit vs. warm path, identical final design	ruled out: no cons
Level-set moves fail	acceptance in the loop vs. in <code>--check</code>	confirmed, but see

Move acceptance is 0.006–0.074 inside the BO loop and exactly 0.000 on clustered designs, against 0.5–0.68 on the spread Sobol designs `--check` uses. Single-site branch flips are almost always rejected once neighbouring points are strongly coupled under the prior, so the sampler stops traversing branches exactly when the design tightens — late in the run, when it matters. The fix is block moves: flip every point within a lengthscale of a random centre, which keeps the proposal symmetric and the prior ratio in closed form.

That is a real defect, but it cannot be the whole story, because **identity** has no branches to move between and still shows a residual.

21.6 The residual on identity

Run with `--no-shortcuts`, **identity** gives “observing h ” = $+0.409 \pm 0.133$ against a true value of 0, with $U_{\text{lost}} = 0.0016$. The sampler recovers h ; the gap is not information.

The remaining asymmetry is how the predictive is formed. The composite arm draws n_{mc} samples from one exact Gaussian conditional. The latent arm spreads the same n_{mc} across a mixture over latent samples, so it covers two sources of randomness with one budget. Raising n_{mc} from 256 to 4096 moves the gap from $+0.409 \pm 0.133$ to $+0.275 \pm 0.168$ — it shrinks, but it does not vanish. About +0.27 is still unaccounted for on a problem where the latent arm provably has complete information.

This is the smallest case that still exhibits the defect: one intermediate, no multimodality, no level-set structure, a posterior that is nearly a point. It is therefore the right place to debug, and everything in Section 21.5 is downstream of it.

21.7 The analogs against the benchmark results

Analog	Benchmark	observing h (corrected)	Composite gain in the paper
product	DTLZ2	+0.213	largest (121%)
max	Lake	+0.139	moderate
ratio	RGB	+0.117	none
norm	Color	−0.095	moderate
sum	Truss, RCM40	−0.180	weakest

The ordering matches at both ends: the DTLZ2 analog pays the most and the Truss/RCM40 analog pays the least, which is what the benchmark study found. Stated carefully, though, these five numbers carry standard errors of 0.06 to 0.25, so no individual pair is separated significantly, and the correction subtracted from all of them is itself uncertain. Treat this as consistent with the benchmark results rather than as confirmation of them.

21.8 Predictions this generates

1. **Local, not global, invertibility should predict the composite gain.** Compute U_{lost} on each real benchmark restricted to near-Pareto-optimal points, and it should correlate with the measured composite gain better than U_{lost} computed over the whole domain. The `relu/clip` contrast predicts this directly and it is cheap to test — no new BO runs, just a posterior fit on the points each existing run already evaluated.
2. **Truss and RCM40 should gain from the extra observations only.** Their outer maps are linear, so knowing g is worth nothing; the whole gain must be the k numbers per evaluation. A direct test is to give the direct arm k evaluations per iteration instead of one — if the gain disappears, it was never about composite structure at all.
3. **A kink at the optimum should not matter, but ambiguity should.** `abs` and `shiftsq_c0` destroy the same information and differ only by smoothness; they measure -0.057 and $+1.490$. If that survives the defects above it says the acquisition function’s difficulty with a kink is a larger effect than the inference problem — worth isolating on its own.
4. **The number of preimages should not be the controlling variable.** $\cos(\omega z)$ at $\omega = 1, 2, 4$ gives $+2.45$, $+1.06$, $+1.08$ — the *fewest* preimages pays the most. If this holds it contradicts the natural reading of “how much information is destroyed” and supports the same local story as the censoring pair.

Part VII

Limitations

- **The hyperparameter asymmetry is not yet removed**, only measurable. See Section 18. “Knowing g ” is an upper bound until the all-MLE run is done.

- **Noise makes injectivity a matter of degree.** As measured in Check 2, an injective g still loses information where its slope is small relative to σ . The clean statement holds only in the noise-free limit.
- **Continuous fibers mix more slowly than discrete ones.** Reflections for z^2 are single moves that jump between modes; following the circle fiber of a norm is a single-site random walk over a smooth field, which mixes more slowly when observed points are strongly coupled. The ratio map has no exact moves at all and relies on elliptical slice sampling alone, so it is the least trustworthy condition in the set.
- **U_{lost} is a variance, not an information measure.** The non-monotonicity at $c = 0.5$ shows the two can come apart when posterior modes are unequally weighted.
- **Nothing here has $k > 2$.** The real benchmarks have four to eight intermediates. Whether the latent arm’s advantage scales with the number of hidden dimensions is untested, and it is also where the sampler would be least reliable, so this is a deliberate omission rather than an oversight.
- **The reference minimum is estimated,** from a large Sobol set refined by gradient ascent, so regret is measured against a very good estimate rather than a proven optimum.
- **Single objective.** The decomposition is cleanest with one objective; whether the same accounting holds once a hypervolume acquisition is involved is a separate question, and the natural next step once these results are in.

A Where each piece lives in the code

Design element	File	Object
Random Fourier feature draws, standardization	test_functions.py	RFFDraw
The nineteen outer maps	test_functions.py	OUTER_MAPS
Level-set moves per map	test_functions.py	OuterMap.moves
Named groups for submission	test_functions.py	GROUPS
Shared prior for arms 2 and 3	arms.py	KnownPrior
Arm 1	arms.py	DirectArm
Arm 2	arms.py	CompositeArm
Arm 3, sampler and shortcuts	arms.py	LatentArm
Elliptical slice step	arms.py	LatentArm._ess_step
Level-set Metropolis moves, Eq. (15)	arms.py	LatentArm._level_set_moves
Projection onto $\{g(h) = y\}$	arms.py	LatentArm._project
Exact Gaussian posterior for linear g	arms.py	exact_linear_posterior
Acquisition	run.py	log_ei, optimize_acq
Paired differences and standard errors	run.py	paired
The three validation checks	run.py	run_checks
Convergence diagnostics	run.py	_mixing
Pooling over function draws	run.py	main, --collect
Cluster submission	cluster/	three_arm.sub, submit_three_arm.sh

References

- [1] I. Murray, R. P. Adams, and D. J. C. MacKay. Elliptical slice sampling. In *Proceedings of the 13th International Conference on Artificial Intelligence and Statistics*, PMLR 9:541–548, 2010.